(iv) $\sum_{n=1}^{102} i^n = \left(i^1+i^2+i^3+i^4\right)+\left(i^5+i^6+i^7+i^8\right)+\cdots+\left(i^{97}+i^{98}+i^{99}+i^{100}\right)+i^{101}+i^{102}$

$= \left(i^1+i^2+i^3+i^4\right)+\left(i^1+i^2+i^3+i^4\right)+\cdots+\left(i^1+i^2+i^3+i^4\right)+i^1+i^2$

$= \{i+(-1)+(-i)+1\}+\{i+(-1)+(-i)+1\}+\cdots\ldots+\{i+(-1)+(-i)+1\}+i+(-1)$

$= 0+0+\cdots 0+i-1$

$= -1+i$ (What is this number?)

(v) $i\, i^2\, i^3 \cdots i^{40} = i^{1+2+3+\cdots+40} = i^{\frac{40\times 41}{2}} = i^{820} = i^0 = 1.$

**Result:** Sum of four consecutive powers of $i$ is zero. That is $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0 \ \forall n \in \mathbb{Z}$

**Note**

(i) $\sqrt{ab} = \sqrt{a}\sqrt{b}$ valid only if at least one of $a$, $b$ is non-negative.

For example, $6 = \sqrt{36} = \sqrt{(-4)(-9)} = \sqrt{(-4)}\sqrt{(-9)} = (2i)(3i) = 6i^2 = -6$, a contradiction.

Since we have taken $\sqrt{(-4)(-9)} = \sqrt{(-4)}\sqrt{(-9)}$, we arrived at a contradiction.

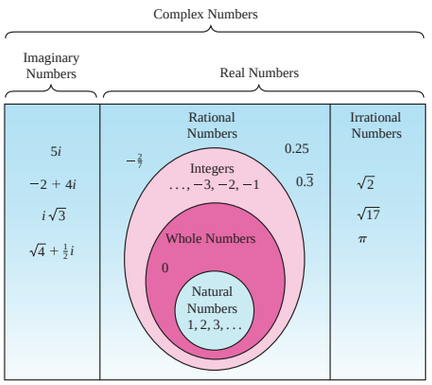
Therefore $\sqrt{ab} = \sqrt{a}\sqrt{b}$ valid only if at least one of $a$, $b$ is non-negative.

(ii) For $y \in \mathbb{R}$, $y^2 \geq 0$

Therefore, $\sqrt{(-1)(y^2)} = \sqrt{(y^2)(-1)}$

$\sqrt{(-1)}\sqrt{(y^2)} = \sqrt{(y^2)}\sqrt{(-1)}$

$iy = yi.$




## EXERCISE 2.1

Simplify the following:

1. $i^{1947} + i^{1950}$
2. $i^{1948} - i^{-1869}$
3. $\sum_{n=1}^{12} i^n$
4. $i^{59} + \frac{1}{i^{59}}$
5. $i\, i^2 i^3 \cdots i^{2000}$
6. $\sum_{n=1}^{10} i^{n+50}$

## 2.2 Complex Numbers

We have seen that the equation $x^2 + 1 = 0$ does not have a solution in real number system.

In general there are polynomial equations with real coefficient which have no real solution.

We enlarge the real number system so as to accommodate solutions of such polynomial equations. This has triggered the mathematicians to define complex number system.

In this section, we define

(i) Complex numbers in rectangular form

(ii) Argand plane

(iii) Algebraic operations on complex numbers

The complex number system is an extension of real number system with imaginary unit $i$.

The imaginary unit $i$ with the property $i^2 = -1$, is combined with two real numbers $x$ and $y$ by the process of addition and multiplication, we obtain a complex number $x + iy$. The symbol '+' is treated as vector addition. It was introduced by Carl Friedrich Gauss (1777-1855).



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